



Engineering Technical Assessment

Background

When pricing products in a grocery store, we consider competition among other factors. We need to index our prices (grocery store A) to those of the grocery stores (B, C, D etc.) that we compete to. To do that, we need to match each of our products to the closest (preferably identical) product sold by each of our competitors.

Exact matches are observed when the exact same product is sold by both grocery stores A and B. Every such product has to be of a national brand (not [private label](#)). Exact matches can be determined using [Universal Product Code \(UPC\)](#) data when available in the datasets of both grocery stores in question. Since we are collecting data online (scraping), this is not always the case.

Non-exact matches are observed for private label and fresh/loose products which either do not have a UPC, or if they do, it is only available within the network of the exact same grocery store. These matches are determined based on all available product attributes such as name, brand, size, form (dry/refrigerated/frozen etc.) and others. Non-exact matches can exist between products of different brands, given that both brands are private label ones. A simple rule of thumb for defining whether two products, each found in a different grocery store, should form a non-exact match is: "would a customer consider both to be essentially the same product?"

Therefore, **product matching** using product attributes other than UPC is necessary in two distinct cases:

- Identifying exact matches when UPC data is not available: Chobani Whole Milk Greek Yogurt Honey Blended 5.3 oz with UPC 0818290011589 is sold by both Walmart ([Walmart product URL](#)) and Wegmans ([Wegmans product URL](#)). Wegmans' online

storefront does not always provide UPC information, preventing us from making that match using UPC.

- Identifying non-exact matches: [Great Value Organic Tomato Sauce 8 oz](#) is essentially the same product as [Wegmans Organic Tomato Sauce 8 oz](#). Both are private label products. We still do not know the UPC of the second product, however, even if we did, it would not be the same as the UPC of the first product. This is a non-exact match, relying on the two products above having identical attributes, essentially being the exact same thing for the customer/consumer of them.

Task

You are given two datasets: products of grocery store A and products of grocery store B, found in this [folder](#). For each product of A, you need to find the **single** closest match from the products available in B.

The deliverables are:

- A list of matches (item_id_A, item_id_B) in CSV
- your executable algorithm in Python that produced the aforementioned list of matches.

The aforementioned list must include **at least 4000 matches** given that the complete set would include more than 10000.

Feel free to make assumptions on how similar do products have to be in order to be considered a non-exact match, same as you would do when shopping in the grocery store.

GPT-5 nano deployment credentials will be provided by BetterBasket to be used in the solution for the above task.